Zahra Rauf

Tech Developer

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Masters in Electrical Engineering (Digital Computing) with hands-on experience in software development, web applications, and teaching programming languages (Java, C, C++, VHDL, Verilog). Skilled in Python, HTML, CSS, SQL, and emerging GenAI tools (LLMs, ML frameworks). Strong communication and problem-solving skills, seeking a junior role in GenAI, backend, or frontend development to drive innovative, high-quality software solutions.

Profiles

Zahra Rauf
GitHub

Zahra Rauf
LinkedIn

Technical Skills

Generative AI
LLMs, ML, Generative Tools, Langchain, RAG Agents, HuggingFace

Prgramming
Python, OOP, JAVA, C, C++, Verilog HDL

Web Development
HTML, CSS, Flask

Backend Development
SQLite, SQLAlchemy

Version Control
Git

Computer Architecture
ARM, ARM Assembly Language, Parallel Processing, Digital Computing, FPGA, ASIC,

Soft Skills

Leadership

Critical Thinking

Creative Approach

Team Player

Certifications

Introduction to JAVA Programming
HKUST
Aug 2015
Java Certification

Languages

English
Bilingual proficiency

Urdu
Native

German
A1

French
Elementary proficiency

Education

Masterschool — **May 2025 - Jan 2026**
Software Engineering — Training Program

- Built and deployed Flask web applications with RESTful APIs, CRUD operations, and SQLAlchemy databases, improving data handling efficiency by 30%.
- Designed and implemented SQL schemas for structured data management and dynamic web functionality, ensuring scalability and reliability.
- Leveraged GitHub for version control, deployment workflows, and comprehensive project documentation, enhancing collaboration and traceability.

University of Engineering & Technology — **Sep 2012 - Aug 2015**
M.Eng in Electrical Engineering — Taxila, Pakistan

- **Specialization**: Digital Computing

University of Engineering & Technology — **Jan 2003 - Mar 2007**
B.Eng in Electrical Engineering — Lahore, Pakistan

- **Specialization**: Computer Engineering

Projects

Urban Regeneration Scenario Engine — **March 2026 - Present**
An AI-assisted MVP exploring district-level regeneration scenarios for Montgomery using municipal data.
Live Demo

• Developed a Smart City scenario engine utilizing municipal open datasets
• Categorized districts based on Business Activity and Urban Stress indicators
• Formulated three unique regeneration strategies for each district, complete with compatibility scoring
• Designed an interactive web map interface for visualizing potential district transformation paths
• Utilized a technology stack including Python, GeoJSON, HTML/CSS/JS, and Render

ProfessorProficient — **Oct 2025 - Present**
GitHub Link

- Developed an AI-driven RAG pipeline utilizing FAISS, OpenAI embeddings, and LangChain, designed to create context-sensitive quizzes from user-submitted documents.
- Devised scalable preprocessing workflows, including chunking, embeddings, and semantic retrieval, to enhance retrieval precision and minimize hallucinations.
- Constructed structured LLM outputs using JSON-enforced prompts and Pydantic validation, ensuring reliable and production-ready quiz generation.

Transform Coefficient Coding Architecture — **Mar 2013 - Aug 2015**
Link to the IEEE Xplore

- While completing my Master's thesis, I engineered a Transform Coefficient Coding Architecture designed to augment the efficiency of the High Efficiency Video Coding (HEVC) standard, which was subsequently evaluated through FPGA Implementation.
- This research was later published in the IEEE Explore Proceedings of ISWSN'17, under the compelling title "Requirement based Transform Coefficient Coding Architecture for DCT/DST for HEVC". (DOI: 10.1109/ISWSN.2017.8250029)

ARM, FPGA, Assembly Language, Verilog HDL, Parallel Processing, HEVC, Video Coding and Processing

MoviWebApp – Movie Management Web Application — **Jun 2025 - Sep 2025**
GitHub Link

- Built and deployed a Flask-based full-stack web app with CRUD functionality for managing movie data, improving data handling efficiency by 25%.
- Designed a responsive front-end using HTML, CSS, and Jinja templates, enhancing user interaction and accessibility.
- Integrated SQLite database and implemented Git/GitHub version control, ensuring reliability, scalability, and seamless deployment on PythonAnywhere.

Experience

Relocation & Personal Development — **Jun 2022 - Oct 2025**

- Transitioned to Germany, focusing on both personal and professional growth.
- Allocated quality time for familial engagements while concurrently pursuing a Software Engineering Certification from Masterschool.

PTUT — **Oct 2020 - May 2022**
Visiting Lecturer — Lahore, Pakistan

- Taught 100+ undergraduate students programming fundamentals in Assembly, VHDL, and Verilog HDL.
- Developed 20+ Control Systems modules, integrating software and hardware design concepts.

NUML — **Feb 2010 – Sep 20I7**
Lecturer — Islamabad, Pakistan

- Conducted in-depth instruction in Java, C++, and HDL (VHDL, Verilog) for more than 200 undergraduate students, attaining an outstanding average evaluation score of 4.5 out of 5.
- Mentored over 20 student research projects in Assembly, Verilog, and C, providing expert guidance in both implementation and debugging stages.